

Factor Analysis

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.956
Bartlett's Test of Sphericity	Approx. Chi-Square	20904.880
	df	595
	Sig.	.000

Communalities

	Initial	Extraction
TEL1	.540	.611
TEL2	.510	.560
TEL3	.576	.650
TEL4	.541	.597
TEL5	.492	.547
FL1	.504	.559
FL2	.540	.621
FL3	.489	.551
FL4	.533	.600
FL5	.515	.582
IDL1	.527	.595
IDL2	.556	.627
IDL3	.479	.541
IDL4	.494	.559
IDL5	.535	.604
EL1	.564	.651
EL2	.501	.553
EL3	.560	.636
EL4	.494	.557
EL5	.494	.546
LA1	.565	.627
LA2	.536	.599
LA3	.547	.618
LA4	.504	.572
LA5	.505	.566
OSE1	.513	.578
OSE2	.478	.528

Communalities

	Initial	Extraction
OSE3	.559	.647
OSE4	.517	.582
OSE5	.520	.583
SPE1	.580	.655
SPE2	.552	.616
SPE3	.510	.564
SPE4	.507	.559
SPE5	.562	.629

Extraction Method: Principal Axis Factoring.

Total Variance Explained

Factor	Initial Eigenvalues			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	12.246	34.989	34.989	11.839	33.826	33.826
2	2.420	6.913	41.902	2.009	5.740	39.567
3	1.995	5.701	47.602	1.585	4.528	44.095
4	1.952	5.577	53.179	1.542	4.407	48.501
5	1.813	5.180	58.360	1.401	4.002	52.503
6	1.624	4.639	62.998	1.219	3.484	55.987
7	1.480	4.228	67.226	1.076	3.073	59.060
8	.538	1.538	68.764			
9	.537	1.534	70.298			
10	.514	1.469	71.767			
11	.512	1.463	73.230			
12	.497	1.419	74.649			
13	.490	1.399	76.048			
14	.477	1.362	77.410			
15	.472	1.348	78.758			
16	.463	1.324	80.082			
17	.444	1.269	81.351			
18	.433	1.239	82.590			
19	.428	1.222	83.812			
20	.427	1.219	85.031			

Total Variance Explained

Factor	Rotation Sums of Squared Loadings ^a
	Total
1	6.577
2	6.974
3	7.306
4	7.432
5	7.171
6	7.950
7	7.963
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	

Total Variance Explained

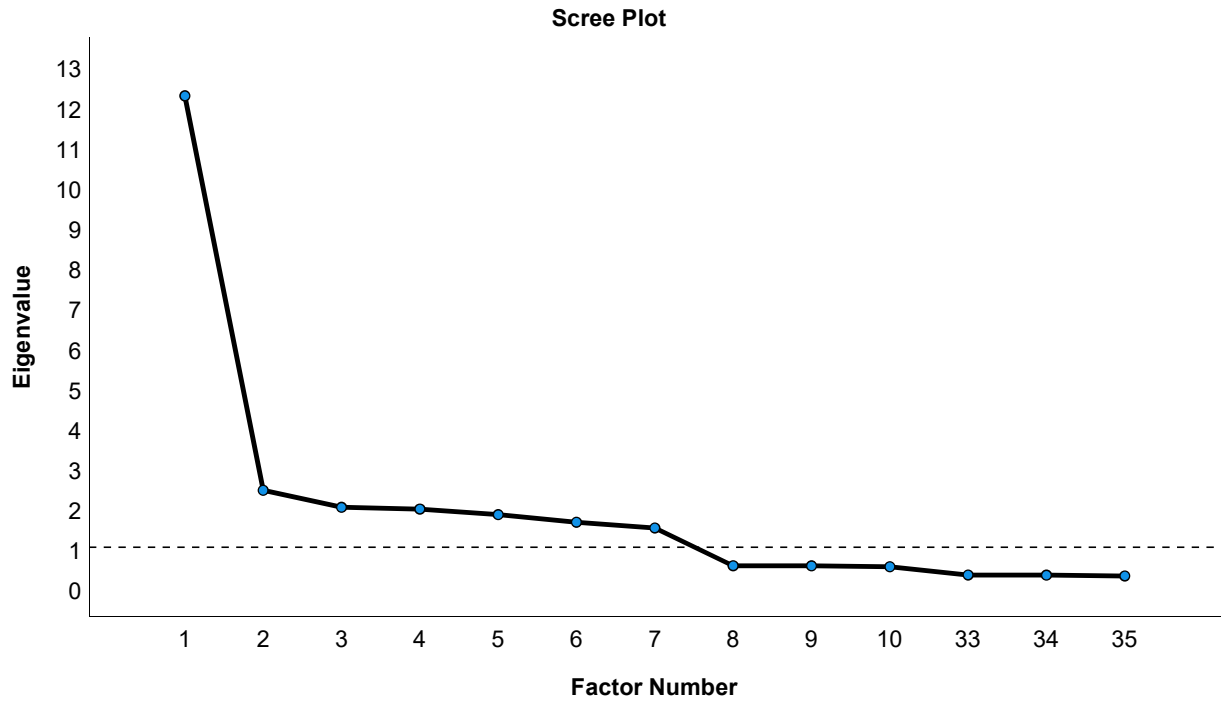
Factor	Initial Eigenvalues			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
21	.411	1.174	86.205			
22	.394	1.124	87.330			
23	.390	1.113	88.443			
24	.381	1.088	89.531			
25	.379	1.083	90.614			
26	.367	1.047	91.661			
27	.359	1.025	92.687			
28	.347	.992	93.678			
29	.346	.989	94.667			
30	.337	.963	95.630			
31	.318	.908	96.538			
32	.313	.894	97.432			
33	.309	.882	98.315			
34	.305	.872	99.187			
35	.285	.813	100.000			

Total Variance Explained

Factor	Rotation Sums of Squared Loadings ^a
	Total
21	
22	
23	
24	
25	
26	
27	
28	
29	
30	
31	
32	
33	
34	
35	

Extraction Method: Principal Axis Factoring.

- a. When factors are correlated, sums of squared loadings cannot be added to obtain a total variance.



Factor Matrix^a

	Factor					
	1	2	3	4	5	6
TEL1	.582					
TEL2	.569					
TEL3	.620					
TEL4	.620					
TEL5	.542					
FL1	.573					
FL2	.575					
FL3	.548					
FL4	.588					
FL5	.551					
IDL1	.557					
IDL2	.567					
IDL3						
IDL4	.515					
IDL5	.530					
EL1	.581					

Factor Matrix^a

	Factor 7
TEL1	
TEL2	
TEL3	
TEL4	
TEL5	
FL1	
FL2	
FL3	
FL4	
FL5	
IDL1	
IDL2	
IDL3	
IDL4	
IDL5	
EL1	

Factor Matrix^a

	Factor					
	1	2	3	4	5	6
EL2	.563					
EL3	.578					
EL4	.516					
EL5	.558					
LA1	.654					
LA2	.622					
LA3	.629					
LA4	.589					
LA5	.601					
OSE1	.567					
OSE2	.560					
OSE3	.601					
OSE4	.578					
OSE5	.575					
SPE1	.643					
SPE2	.627					
SPE3	.599					
SPE4	.610					
SPE5	.634					

Factor Matrix^a

	Factor 7
EL2	
EL3	
EL4	
EL5	
LA1	
LA2	
LA3	
LA4	
LA5	
OSE1	
OSE2	
OSE3	
OSE4	
OSE5	
SPE1	
SPE2	
SPE3	
SPE4	
SPE5	

Extraction Method: Principal Axis Factoring.

a. 7 factors extracted. 5 iterations required.

Pattern Matrix^a

	Factor					
	1	2	3	4	5	6
TEL1				.793		
TEL2				.743		
TEL3				.789		
TEL4				.722		
TEL5				.763		
FL1					.710	
FL2					.800	
FL3					.745	
FL4					.743	
FL5					.782	
IDL1	.746					
IDL2	.770					
IDL3	.750					
IDL4	.749					
IDL5	.774					
EL1		.806				
EL2		.711				
EL3		.798				
EL4		.773				
EL5		.707				
LA1						
LA2						
LA3						
LA4						
LA5						
OSE1			.754			
OSE2			.706			
OSE3			.815			
OSE4			.746			
OSE5			.742			
SPE1						.806
SPE2						.770
SPE3						.746

Pattern Matrix^a

	Factor 7
TEL1	
TEL2	
TEL3	
TEL4	
TEL5	
FL1	
FL2	
FL3	
FL4	
FL5	
IDL1	
IDL2	
IDL3	
IDL4	
IDL5	
EL1	
EL2	
EL3	
EL4	
EL5	
LA1	.748
LA2	.754
LA3	.766
LA4	.771
LA5	.750
OSE1	
OSE2	
OSE3	
OSE4	
OSE5	
SPE1	
SPE2	
SPE3	

Pattern Matrix^a

	Factor					
	1	2	3	4	5	6
SPE4						.717
SPE5						.780

Pattern Matrix^a

	Factor
	7
SPE4	
SPE5	

Extraction Method: Principal Axis Factoring.

Rotation Method: Promax with Kaiser Normalization.^a

a. Rotation converged in 6 iterations.

Structure Matrix

	Factor					
	1	2	3	4	5	6
TEL1				.780		
TEL2				.746		
TEL3				.805		
TEL4				.770		
TEL5				.737		
FL1					.743	
FL2					.788	
FL3					.741	
FL4					.772	
FL5					.762	
IDL1	.770					
IDL2	.791					
IDL3	.734					
IDL4	.744					
IDL5	.774					
EL1		.806				
EL2		.742				
EL3		.795				
EL4		.745				

Structure Matrix

	Factor 7
TEL1	
TEL2	
TEL3	
TEL4	
TEL5	
FL1	
FL2	
FL3	
FL4	
FL5	
IDL1	
IDL2	
IDL3	
IDL4	
IDL5	
EL1	
EL2	
EL3	
EL4	

Structure Matrix

	Factor					
	1	2	3	4	5	6
EL5		.737				
LA1						
LA2						
LA3						
LA4						
LA5						
OSE1			.759			
OSE2			.726			
OSE3			.804			
OSE4			.762			
OSE5			.762			
SPE1						.809
SPE2						.784
SPE3						.751
SPE4						.747
SPE5						.793

Structure Matrix

	Factor 7
EL5	
LA1	.789
LA2	.772
LA3	.783
LA4	.754
LA5	.750
OSE1	
OSE2	
OSE3	
OSE4	
OSE5	
SPE1	
SPE2	
SPE3	
SPE4	
SPE5	

Extraction Method: Principal Axis Factoring.

Rotation Method: Promax with Kaiser Normalization.

Factor Correlation Matrix

Factor	1	2	3	4	5	6
1	1.000	.466	.376	.438	.472	.461
2	.466	1.000	.429	.474	.483	.496
3	.376	.429	1.000	.481	.455	.599
4	.438	.474	.481	1.000	.512	.528
5	.472	.483	.455	.512	1.000	.466
6	.461	.496	.599	.528	.466	1.000
7	.475	.479	.565	.540	.517	.569

Factor Correlation Matrix

Factor	7
1	.475
2	.479
3	.565
4	.540
5	.517
6	.569
7	1.000

Extraction Method: Principal Axis Factoring.

Rotation Method: Promax with Kaiser Normalization.

Multilayer Perceptron

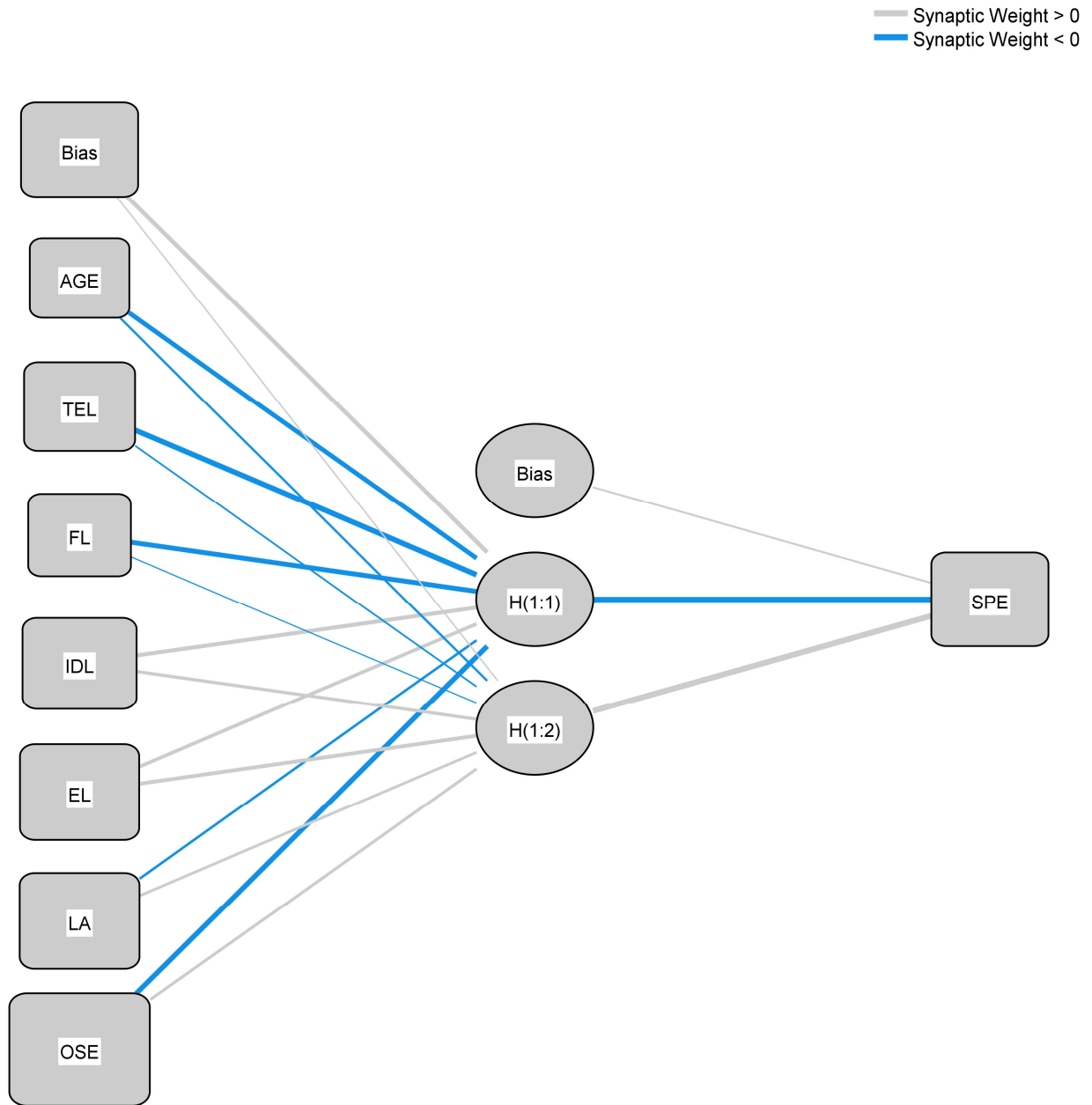
Case Processing Summary

		N	Percent
Sample	Training	773	70.3%
	Testing	327	29.7%
Valid		1100	100.0%
Excluded		0	
Total		1100	

Network Information

Input Layer	Covariates	1	AGE
		2	TEL
		3	FL
		4	IDL
		5	EL
		6	LA
		7	OSE
	Number of Units ^a		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		2
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	SPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

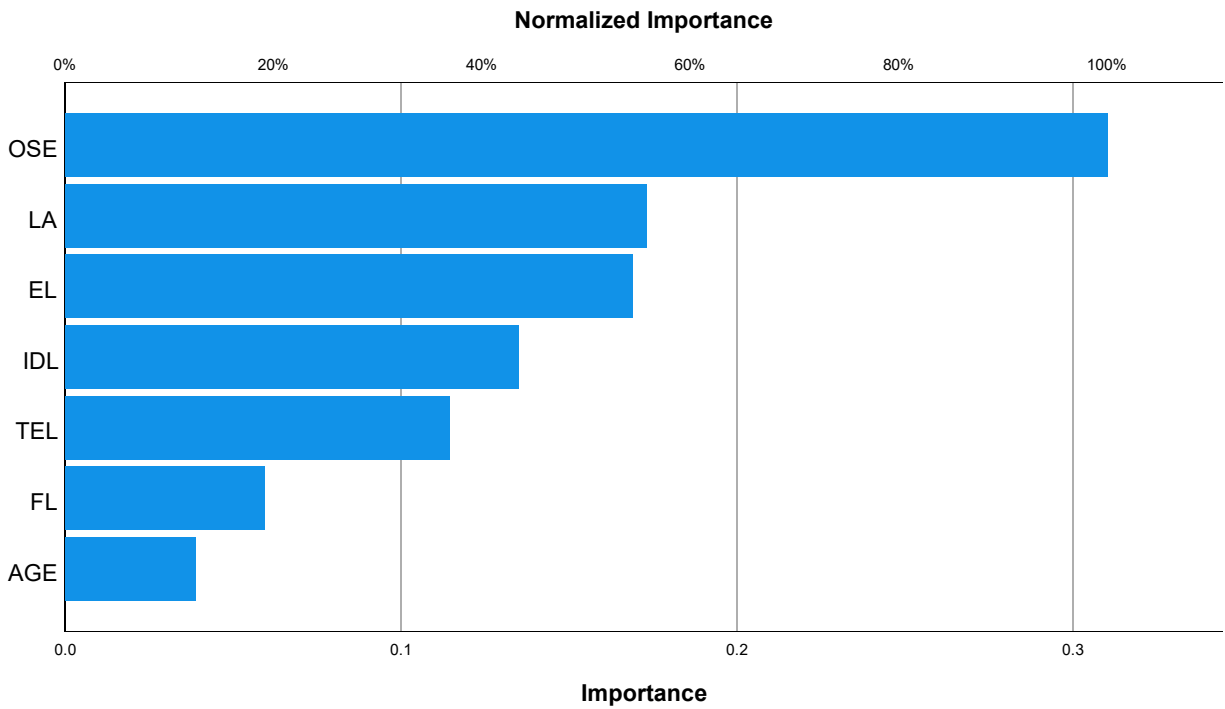
Training	Sum of Squares Error	212.074
	Relative Error	.549
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.10
Testing	Sum of Squares Error	95.073
	Relative Error	.584

Dependent Variable: SPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
AGE	.039	12.5%
TEL	.115	36.9%
FL	.059	19.1%
IDL	.135	43.5%
EL	.169	54.5%
LA	.173	55.8%
OSE	.310	100.0%



Multilayer Perceptron

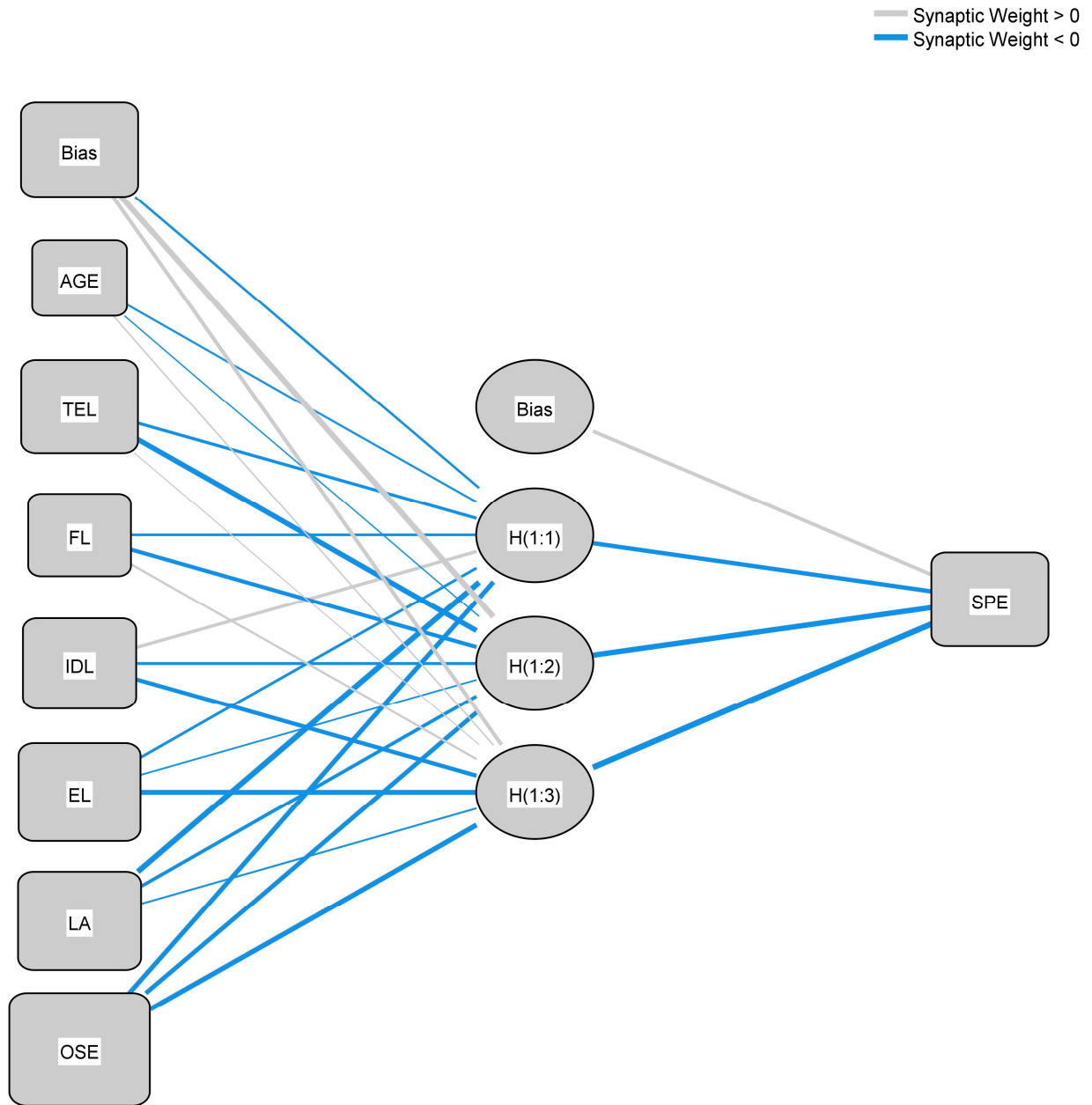
Case Processing Summary

		N	Percent
Sample	Training	763	69.4%
	Testing	337	30.6%
Valid		1100	100.0%
Excluded		0	
Total		1100	

Network Information

Input Layer	Covariates	1	AGE
		2	TEL
		3	FL
		4	IDL
		5	EL
		6	LA
		7	OSE
	Number of Units ^a		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		3
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	SPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent
Output layer activation function: Identity

Model Summary

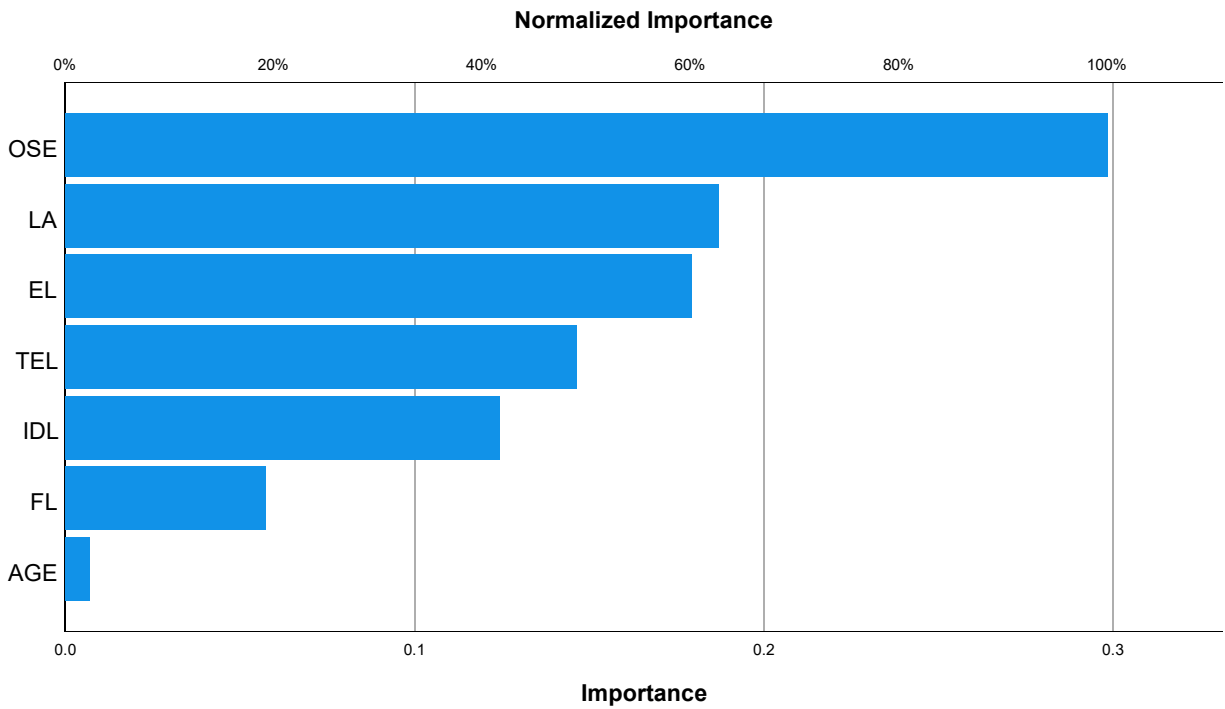
Training	Sum of Squares Error	219.274
	Relative Error	.576
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.18
Testing	Sum of Squares Error	100.167
	Relative Error	.548

Dependent Variable: SPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
AGE	.007	2.3%
TEL	.146	49.1%
FL	.057	19.3%
IDL	.124	41.7%
EL	.179	60.1%
LA	.187	62.7%
OSE	.298	100.0%



Multilayer Perceptron

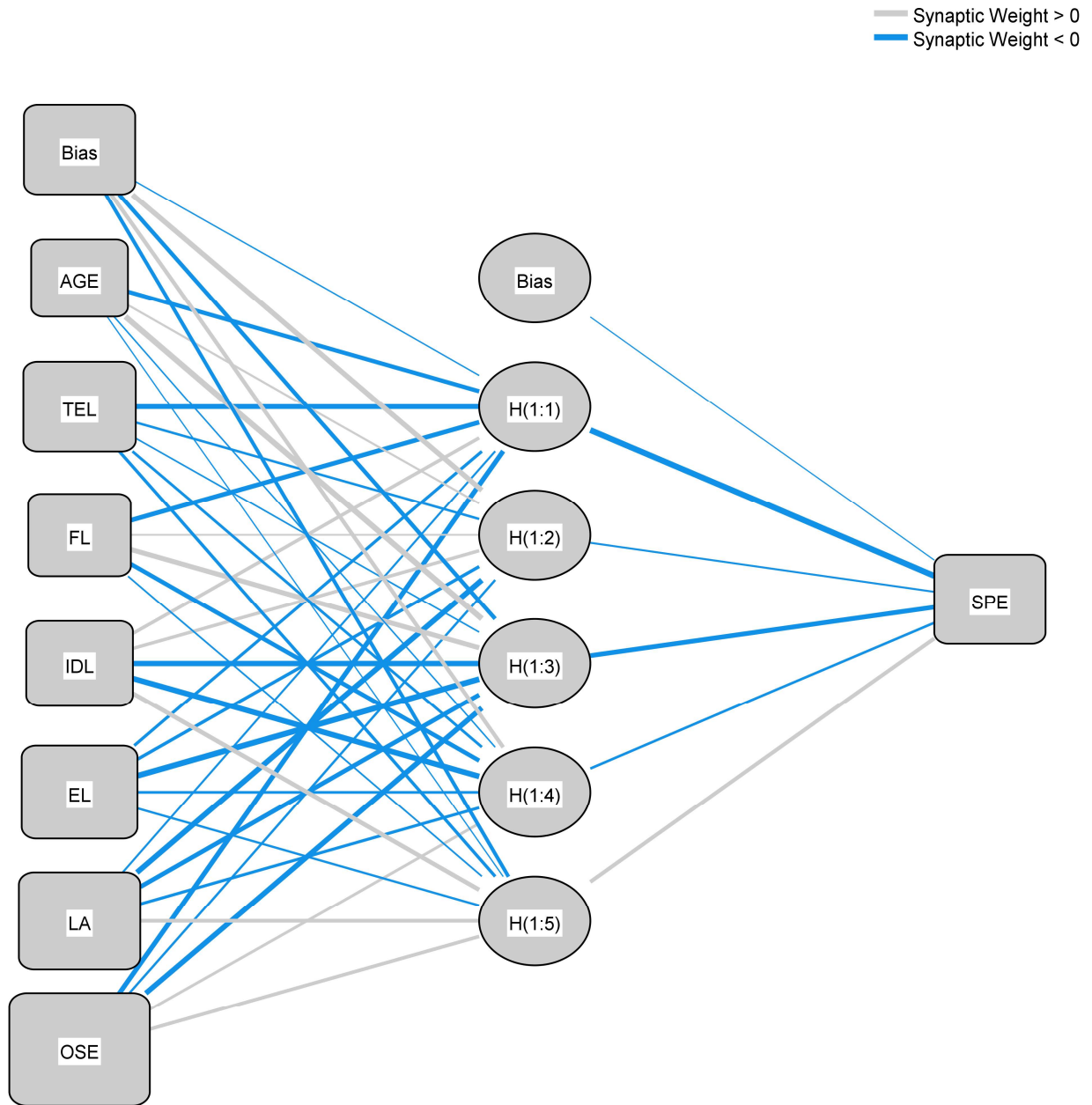
Case Processing Summary

		N	Percent
Sample	Training	757	68.8%
	Testing	343	31.2%
Valid		1100	100.0%
Excluded		0	
Total		1100	

Network Information

Input Layer	Covariates	1	AGE
		2	TEL
		3	FL
		4	IDL
		5	EL
		6	LA
		7	OSE
	Number of Units ^a		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		5
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	SPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

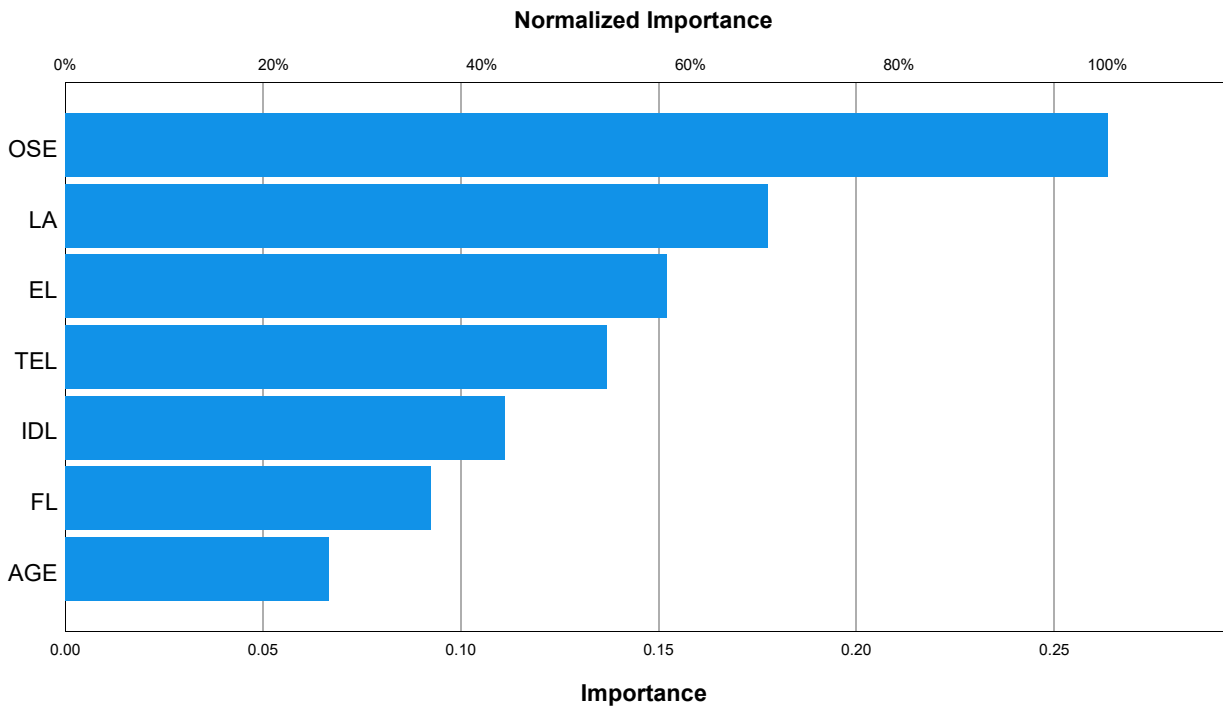
Training	Sum of Squares Error	212.212
	Relative Error	.561
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.10
Testing	Sum of Squares Error	98.889
	Relative Error	.593

Dependent Variable: SPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
AGE	.066	25.2%
TEL	.137	52.0%
FL	.092	35.1%
IDL	.111	42.2%
EL	.152	57.7%
LA	.178	67.4%
OSE	.263	100.0%



Multilayer Perceptron

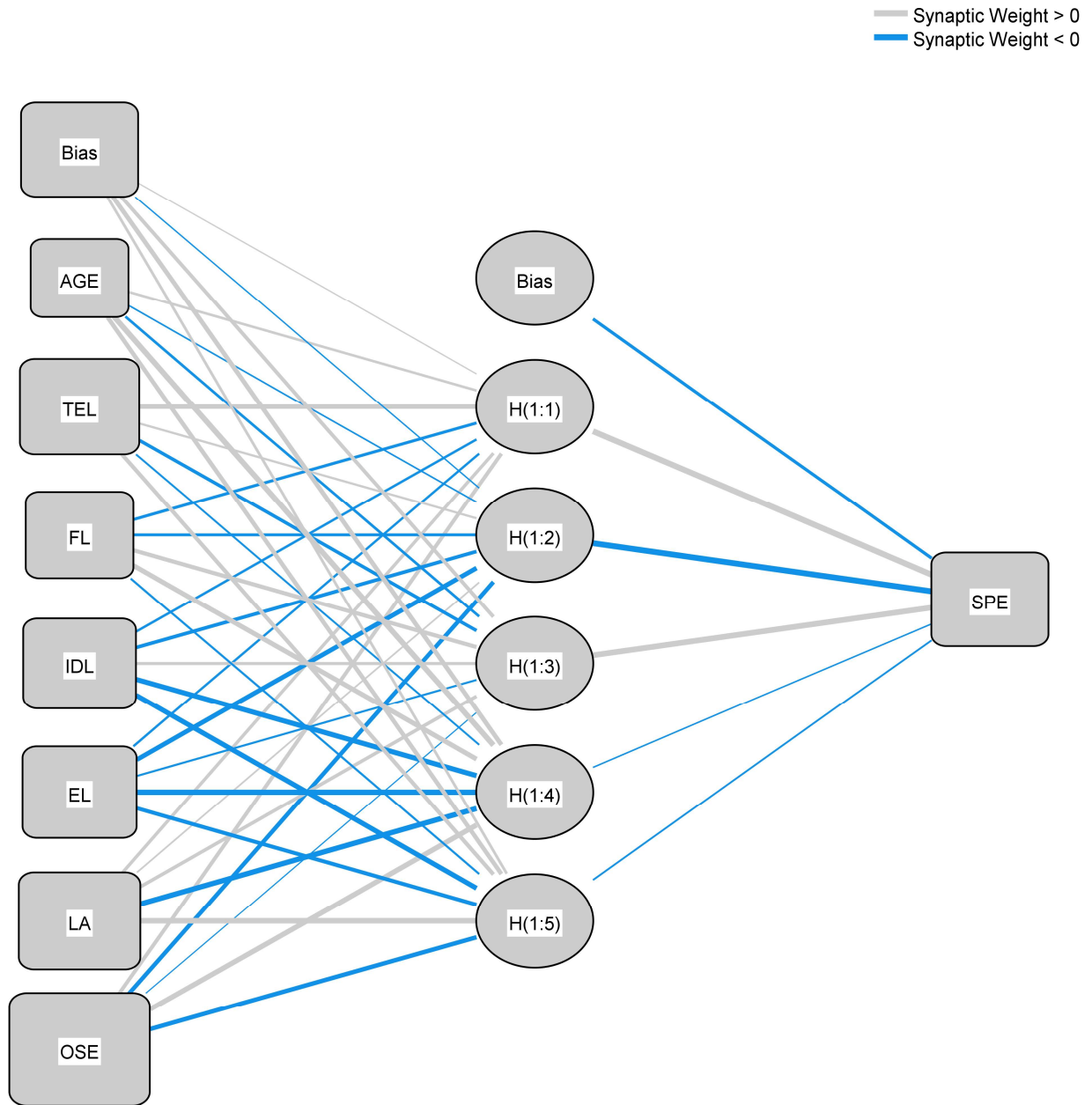
Case Processing Summary

		N	Percent
Sample	Training	776	70.5%
	Testing	324	29.5%
Valid		1100	100.0%
Excluded		0	
Total		1100	

Network Information

Input Layer	Covariates	1	AGE
		2	TEL
		3	FL
		4	IDL
		5	EL
		6	LA
		7	OSE
	Number of Units ^a		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		5
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	SPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent
Output layer activation function: Identity

Model Summary

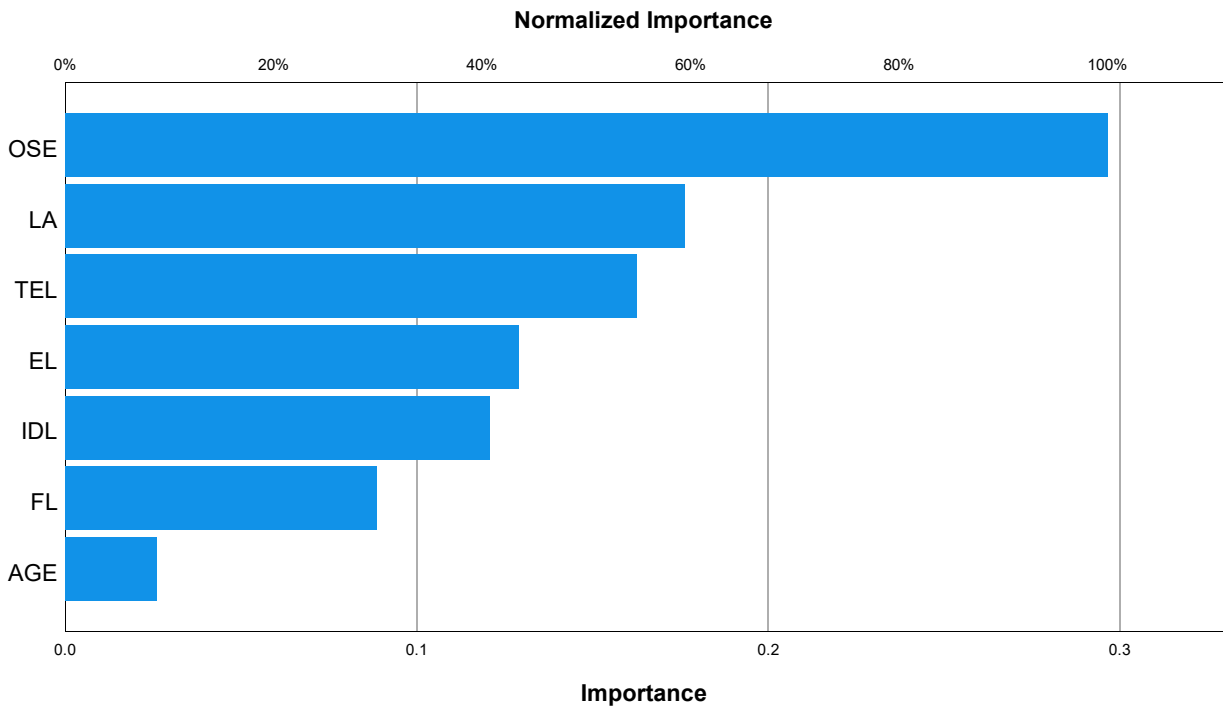
Training	Sum of Squares Error	215.192
	Relative Error	.555
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.16
Testing	Sum of Squares Error	89.046
	Relative Error	.579

Dependent Variable: SPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
AGE	.026	8.8%
TEL	.163	54.9%
FL	.089	29.9%
IDL	.121	40.7%
EL	.129	43.5%
LA	.176	59.5%
OSE	.296	100.0%



Multilayer Perceptron

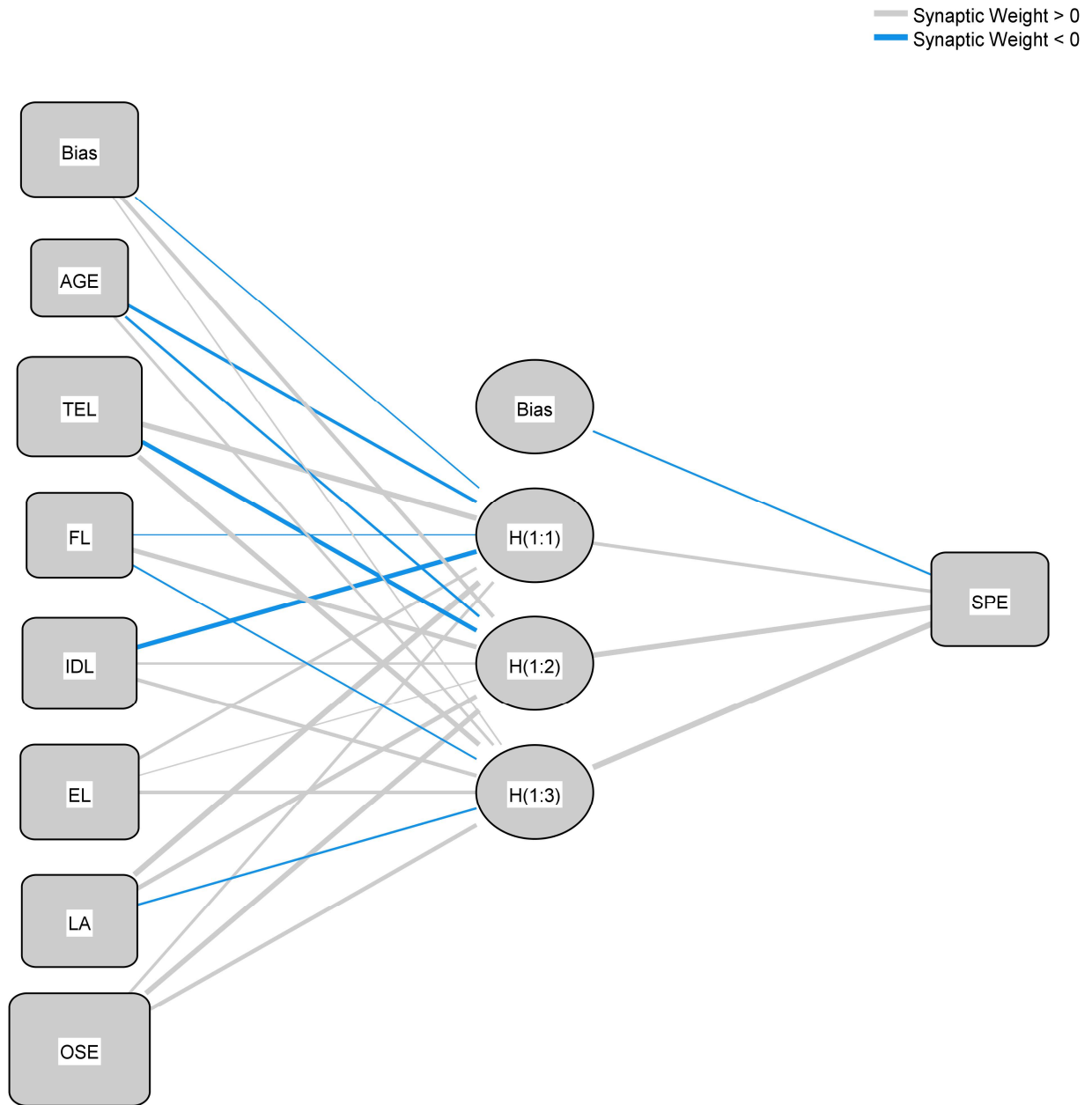
Case Processing Summary

		N	Percent
Sample	Training	763	69.4%
	Testing	337	30.6%
Valid		1100	100.0%
Excluded		0	
Total		1100	

Network Information

Input Layer	Covariates	1	AGE
		2	TEL
		3	FL
		4	IDL
		5	EL
		6	LA
		7	OSE
	Number of Units ^a		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		3
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	SPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent
Output layer activation function: Identity

Model Summary

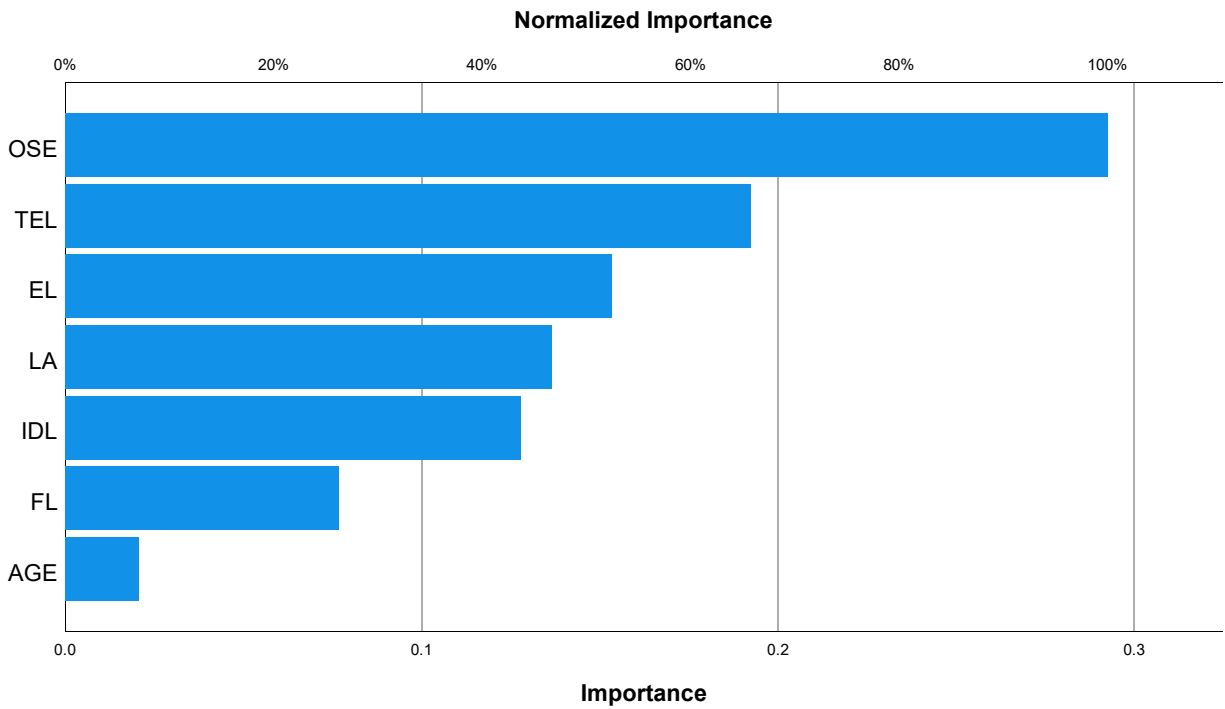
Training	Sum of Squares Error	213.127
	Relative Error	.559
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.09
Testing	Sum of Squares Error	94.964
	Relative Error	.589

Dependent Variable: SPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
AGE	.021	7.0%
TEL	.192	65.8%
FL	.077	26.2%
IDL	.128	43.7%
EL	.154	52.5%
LA	.136	46.6%
OSE	.292	100.0%



Multilayer Perceptron

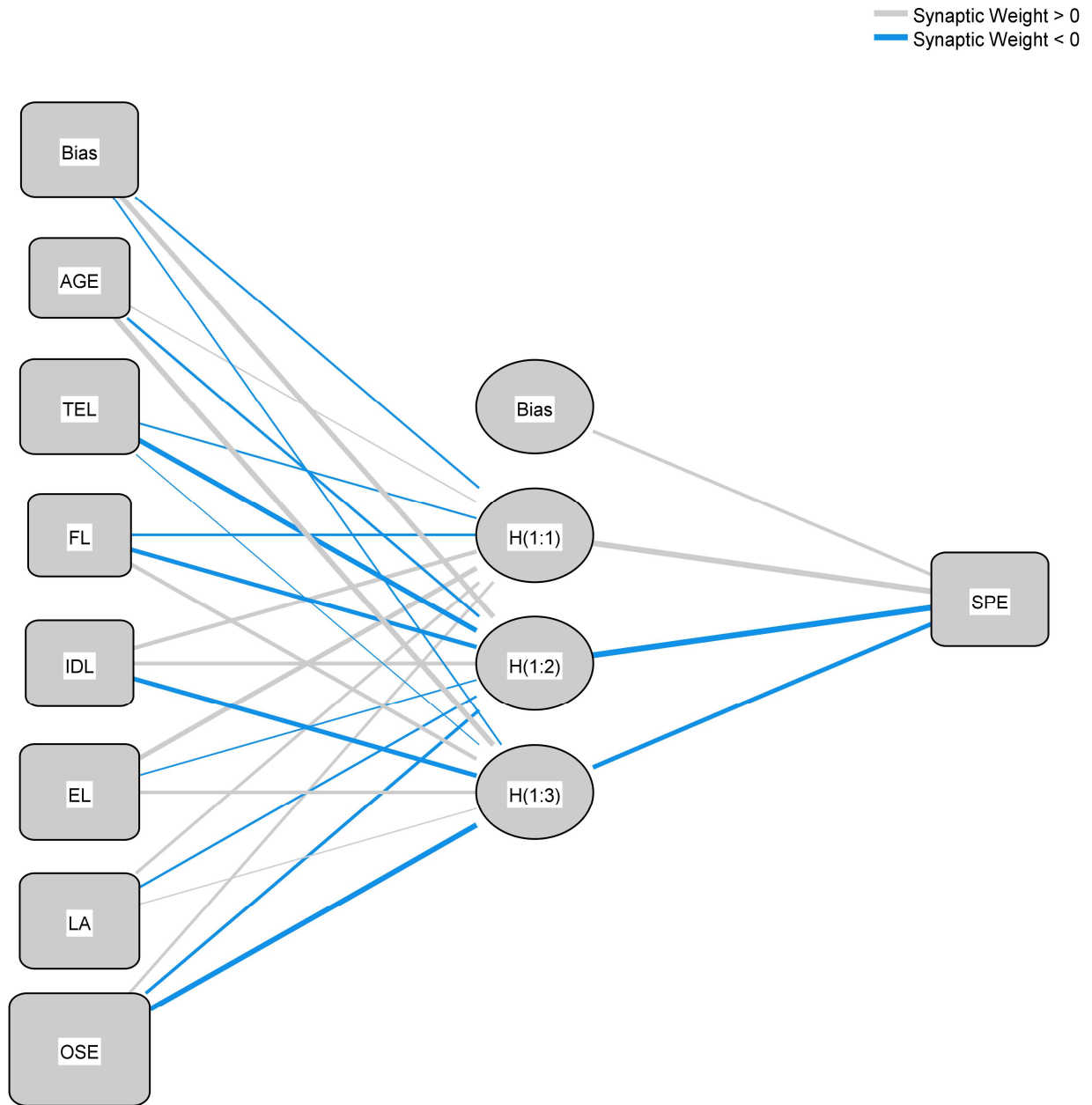
Case Processing Summary

		N	Percent
Sample	Training	771	70.1%
	Testing	329	29.9%
Valid		1100	100.0%
Excluded		0	
Total		1100	

Network Information

Input Layer	Covariates	1	AGE
		2	TEL
		3	FL
		4	IDL
		5	EL
		6	LA
		7	OSE
	Number of Units ^a		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		3
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	SPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

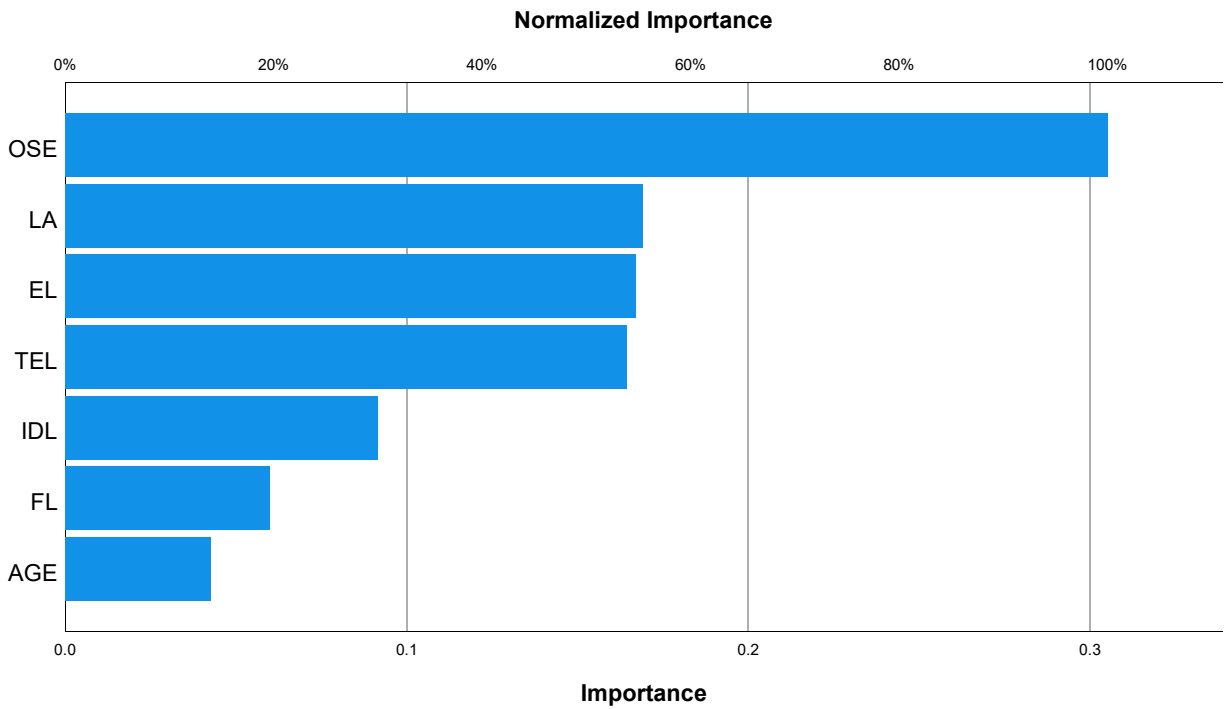
Training	Sum of Squares Error	213.817
	Relative Error	.555
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.10
Testing	Sum of Squares Error	82.936
	Relative Error	.578

Dependent Variable: SPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
AGE	.043	14.0%
TEL	.164	53.9%
FL	.060	19.6%
IDL	.092	30.0%
EL	.167	54.8%
LA	.169	55.4%
OSE	.305	100.0%



Multilayer Perceptron

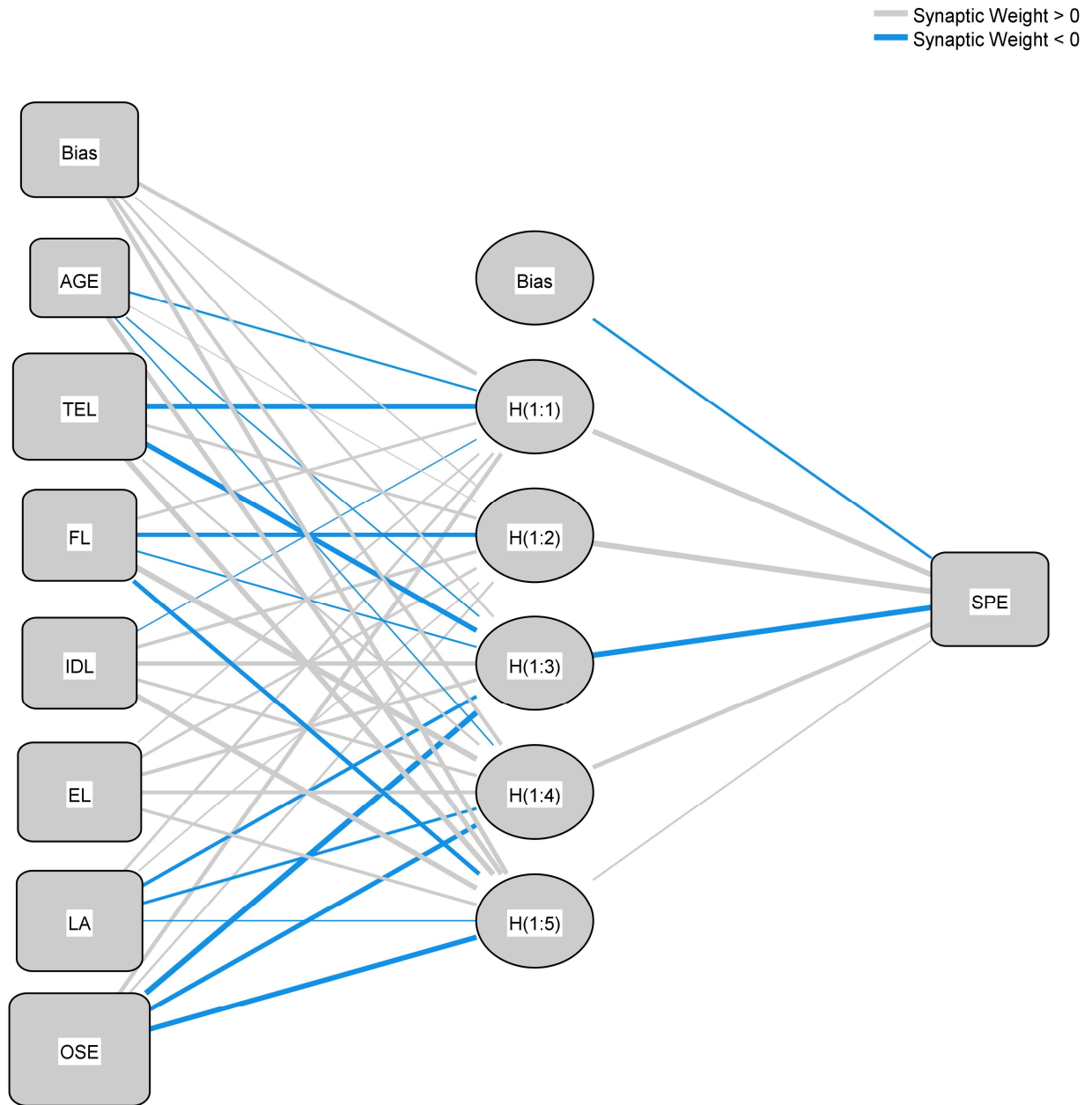
Case Processing Summary

		N	Percent
Sample	Training	766	69.6%
	Testing	334	30.4%
Valid		1100	100.0%
Excluded		0	
Total		1100	

Network Information

Input Layer	Covariates	1	AGE
		2	TEL
		3	FL
		4	IDL
		5	EL
		6	LA
		7	OSE
	Number of Units ^a		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		5
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	SPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

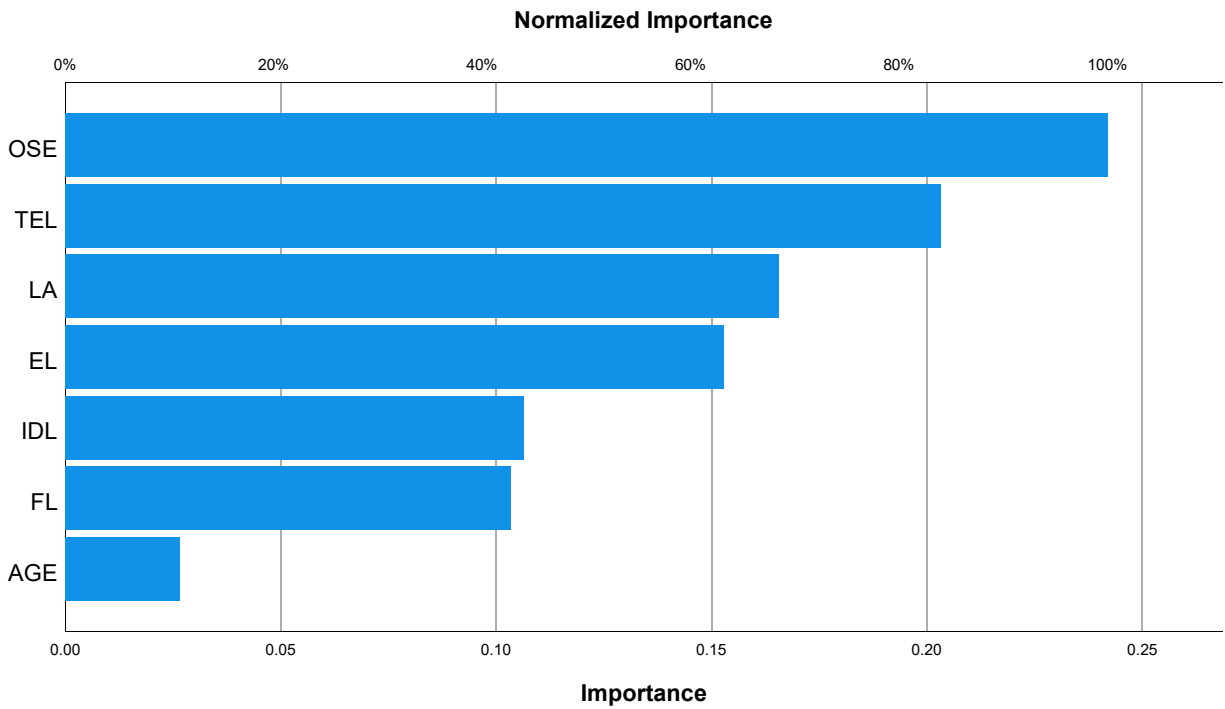
Training	Sum of Squares Error	204.405
	Relative Error	.534
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.15
Testing	Sum of Squares Error	92.449
	Relative Error	.629

Dependent Variable: SPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
AGE	.027	11.0%
TEL	.203	84.0%
FL	.103	42.8%
IDL	.106	44.0%
EL	.153	63.2%
LA	.166	68.5%
OSE	.242	100.0%



Multilayer Perceptron

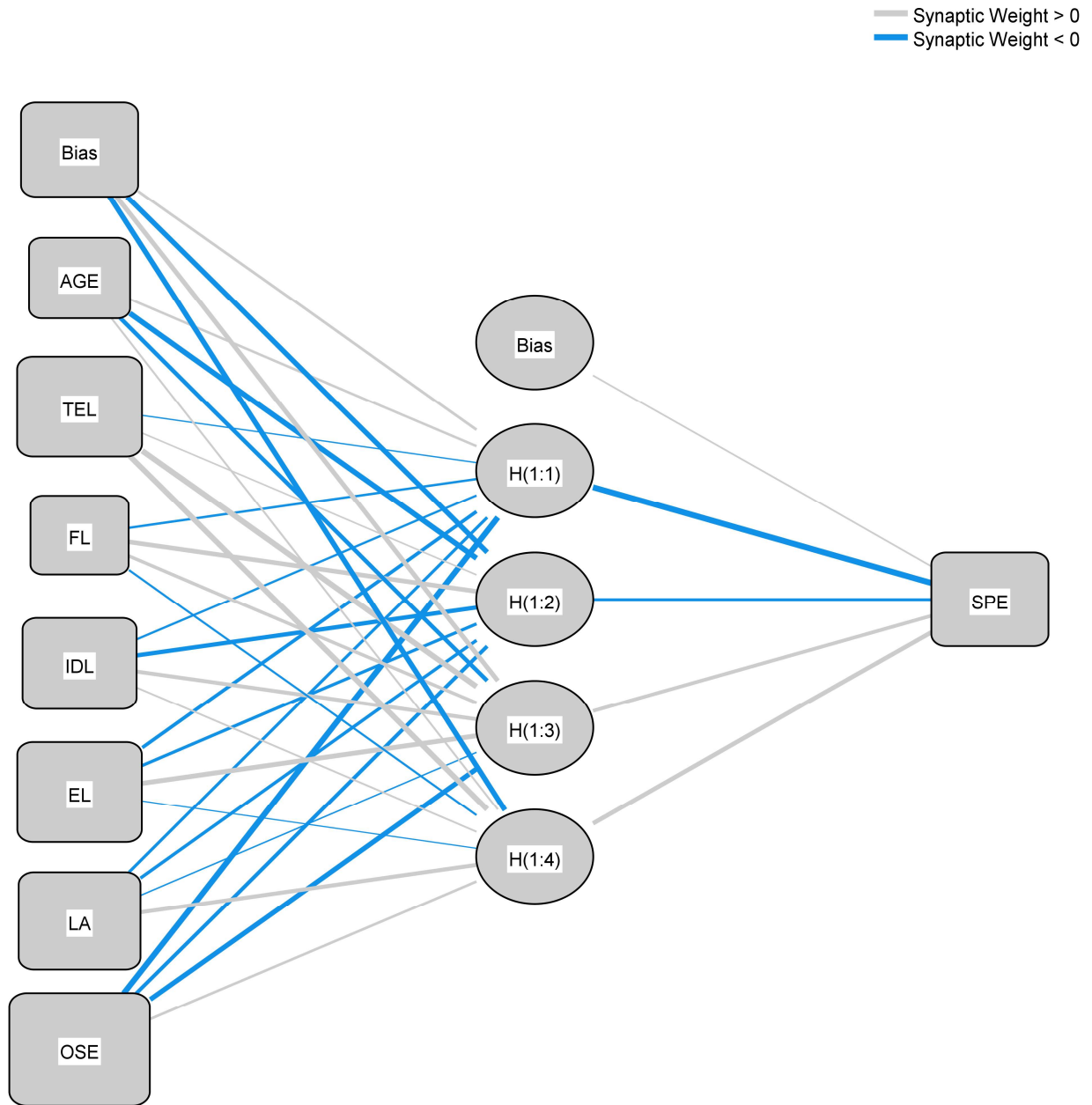
Case Processing Summary

		N	Percent
Sample	Training	775	70.5%
	Testing	325	29.5%
Valid		1100	100.0%
Excluded		0	
Total		1100	

Network Information

Input Layer	Covariates	1	AGE
		2	TEL
		3	FL
		4	IDL
		5	EL
		6	LA
		7	OSE
	Number of Units ^a		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		4
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	SPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

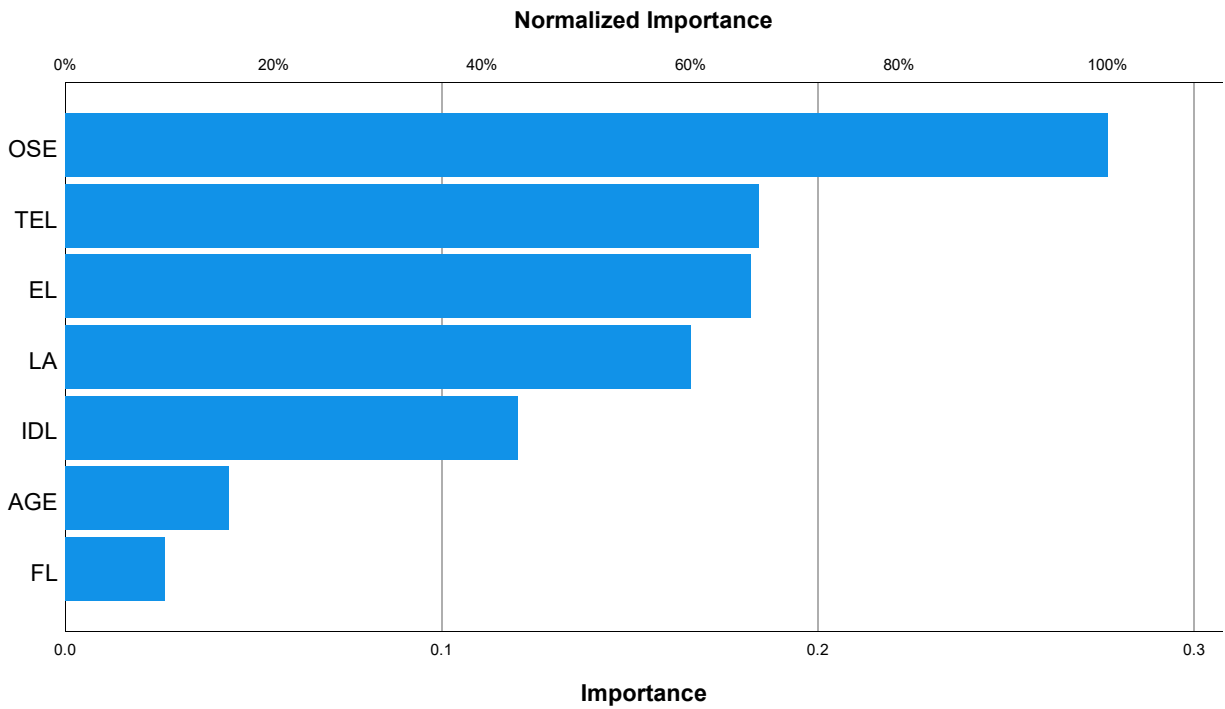
Training	Sum of Squares Error	230.307
	Relative Error	.595
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.10
Testing	Sum of Squares Error	76.387
	Relative Error	.502

Dependent Variable: SPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
AGE	.043	15.7%
TEL	.184	66.5%
FL	.027	9.6%
IDL	.120	43.5%
EL	.182	65.8%
LA	.166	60.0%
OSE	.277	100.0%



Multilayer Perceptron

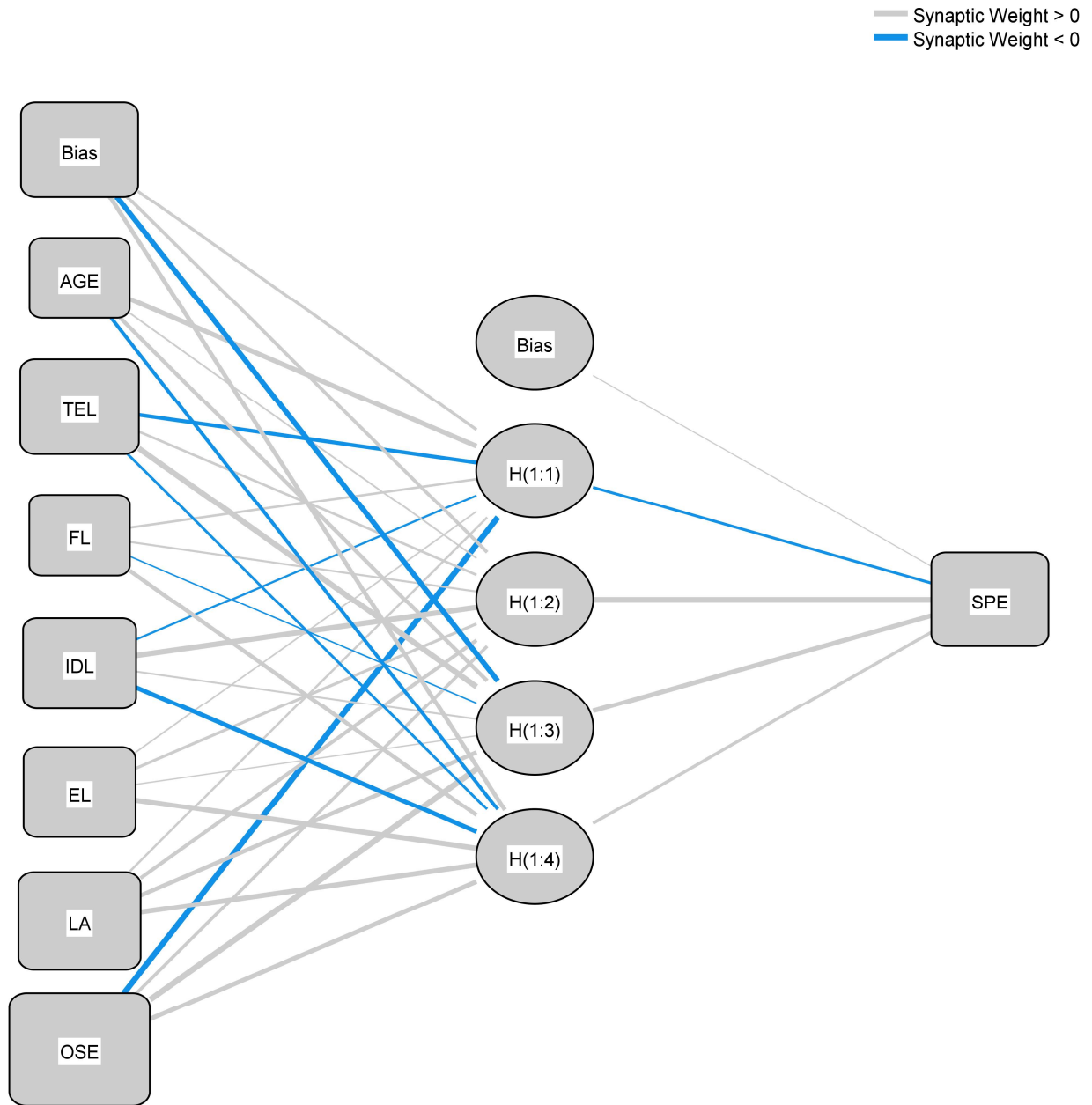
Case Processing Summary

		N	Percent
Sample	Training	780	70.9%
	Testing	320	29.1%
Valid		1100	100.0%
Excluded		0	
Total		1100	

Network Information

Input Layer	Covariates	1	AGE
		2	TEL
		3	FL
		4	IDL
		5	EL
		6	LA
		7	OSE
	Number of Units ^a		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		4
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	SPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

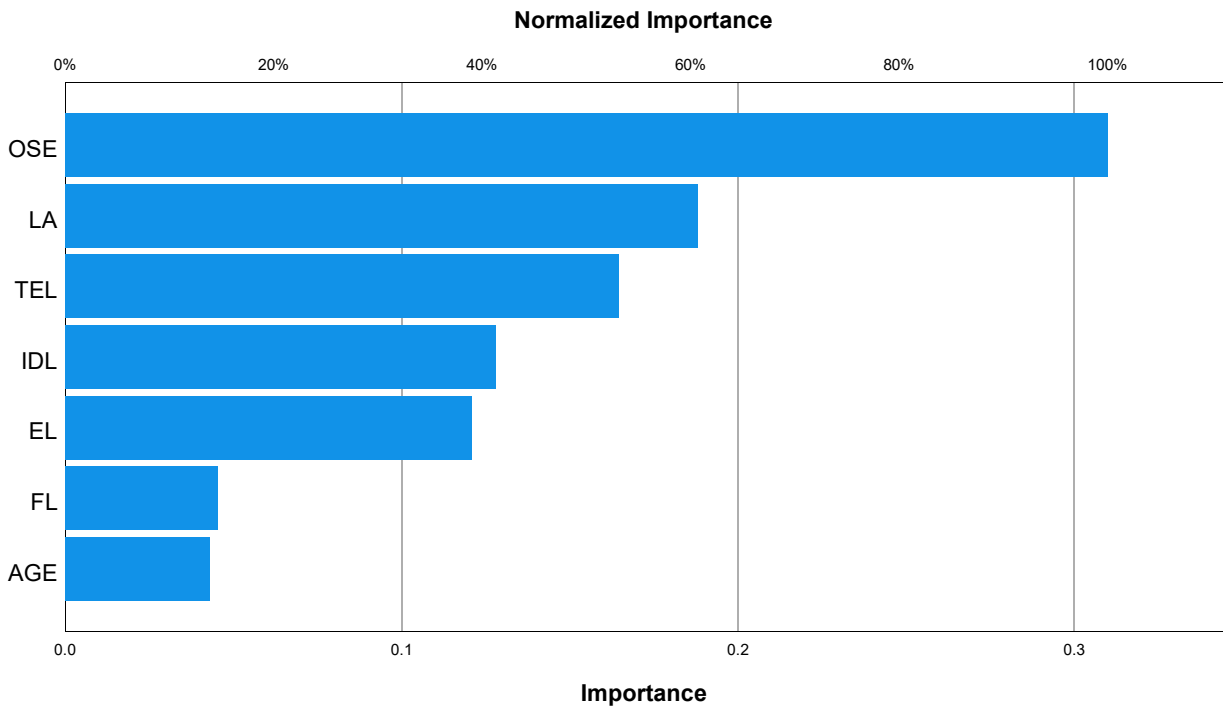
Training	Sum of Squares Error	230.368
	Relative Error	.591
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.11
Testing	Sum of Squares Error	81.676
	Relative Error	.520

Dependent Variable: SPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
AGE	.043	13.9%
TEL	.165	53.1%
FL	.046	14.7%
IDL	.128	41.3%
EL	.121	39.0%
LA	.188	60.7%
OSE	.310	100.0%



Multilayer Perceptron

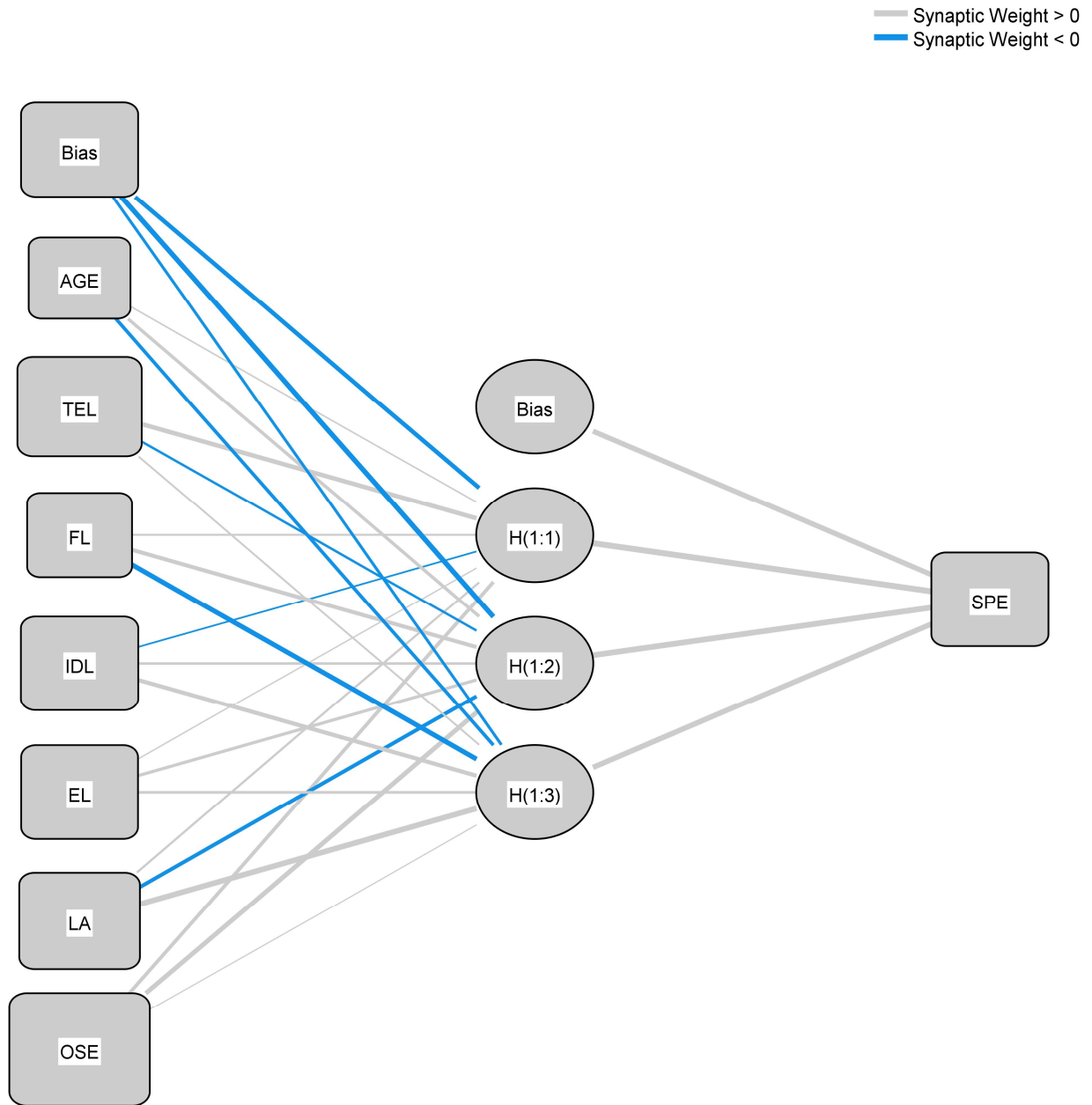
Case Processing Summary

		N	Percent
Sample	Training	777	70.6%
	Testing	323	29.4%
Valid		1100	100.0%
Excluded		0	
Total		1100	

Network Information

Input Layer	Covariates	1	AGE
		2	TEL
		3	FL
		4	IDL
		5	EL
		6	LA
		7	OSE
	Number of Units ^a		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		3
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	SPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

Training	Sum of Squares Error	224.261
	Relative Error	.578
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.12
Testing	Sum of Squares Error	80.065
	Relative Error	.508

Dependent Variable: SPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
AGE	.047	17.3%
TEL	.179	65.4%
FL	.066	23.9%
IDL	.139	50.6%
EL	.136	49.4%
LA	.158	57.6%
OSE	.275	100.0%

